

Analysis of Deep Learning Architectures for Real-Time Facial Recognition

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Abstract

This study deals with the real-time recognition of people using CNN-based deep learning models. It reviews and compares four different architectures: LightCNN-9, [Architecture 2], [Architecture 3], and [Architecture 4]. The comparison uses the experimental results and figures reported in the original research papers. Wherever possible, results obtained using the same facial dataset and evaluation protocol are selected to maintain a fair comparison. The models are compared based on recognition accuracy, computational requirements, inference speed, and suitability for real-time use. The study aims to identify which architecture provides the best balance between recognition performance and computational efficiency for a real-time facial recognition system. Later, the best-performing model will be used to build a real-time face recognition system using our own dataset.

I. Introduction

Facial recognition is a domain of computer vision that deals with recognising a human face and assigning a label or identity to it. It is commonly used in security systems, access control, attendance monitoring, surveillance, mobile-device authentication, and the organisation of digital images. Unlike face detection, which only finds the location of a face within an image, facial recognition attempts to determine the identity of the detected person. In a real-time facial recognition system, this complete process must be performed quickly enough to recognise people from a live image or video stream. Therefore, such a system requires not only good recognition accuracy but also efficient processing and a suitable response time.

The major challenge in facial recognition lies in the large variety of human faces and the different

conditions in which facial images can be captured. Changes in the surrounding environment, lighting, facial position, camera angle, facial expression, and distance from the camera can significantly affect recognition performance. Other factors, such as low image resolution, image size, partial facial obstruction, background variation, ageing, and poor camera quality, can make the task even more difficult. The same person may appear different under different conditions, while two different people may sometimes have similar facial features. A reliable facial recognition model must therefore learn the important features that represent a person's identity while reducing the effect of these unwanted variations.

Within this field, Convolutional Neural Networks (CNNs) have become one of the major deep learning models used for recognising and classifying images. CNNs learn features directly from image data through hierarchical representations. The earlier layers generally learn simple features such as edges, lines, and textures, while the deeper layers learn more complex patterns such as facial structures and identity-related features. Their local connectivity and shared-weight mechanism allow them to process images efficiently and make them highly versatile for different image-related tasks. However, different CNN architectures may produce different results in terms of recognition accuracy, model size, computational requirements, and processing speed.

In this study, four CNN-based architectures are reviewed and compared for facial recognition: LightCNN-9, [Architecture 2], [Architecture 3], and [Architecture 4]. LightCNN-9 is a compact facial recognition architecture that uses the Max-Feature-Map activation to select more informative features and reduce the effect of less useful information. Its relatively lightweight

structure makes it an important candidate for a real-time facial recognition system where computational efficiency is required. The remaining architectures are also studied to understand how their designs influence recognition performance, efficiency, and suitability for real-time use.

The comparison is based on the experimental results and figures reported in the original research papers. To make the comparison as fair as possible, results obtained using similar datasets, evaluation protocols, preprocessing methods, and performance metrics are selected wherever they are available. Results from different testing conditions are not treated as directly equivalent, and these differences are considered while discussing the performance of each architecture. The models are compared using factors such as recognition accuracy, inference speed, model size, and computational requirements.

The main purpose of this study is to identify the architecture that provides the best balance between recognition performance and computational efficiency. The best-performing and most suitable model will later be selected to build a real-time facial recognition system trained on our own dataset. This comparison will therefore provide the foundation for choosing a model that can recognise the required individuals accurately while still being practical for real-time implementation.

II. Architecture 1: LightCNN-9

LightCNN was introduced by Wu et al. to learn facial features from large-scale datasets containing incorrectly labelled and noisy images [1]. The main objective of this architecture is to achieve reliable facial recognition while reducing the model size and computational requirements. LightCNN-9 is one of the smaller architectures proposed in the original study, making it suitable for systems where recognition speed and available hardware are important.

The main feature of LightCNN is the Max-Feature-Map (MFM) operation, which is used instead of a conventional activation function such as ReLU. A convolutional layer first produces $2N$ feature maps, which are divided into pairs.

MFM compares the corresponding values in each pair and keeps the maximum value:

$$\hat{x}_{ij}^k = \max(x_{ij}^k, x_{ij}^{k+N}) \quad (1)$$

As a result, $2N$ input feature maps are reduced to N output feature maps. This creates competition between the feature maps and allows the network to retain the stronger facial information while suppressing weaker or noisy responses.

LightCNN-9 takes a 128×128 grayscale facial image as its input. The first stage contains a 5×5 convolution followed by MFM and max pooling. It is followed by four convolution blocks, with each block containing a 1×1 convolution, MFM, a 3×3 convolution, and another MFM operation. Four max-pooling operations gradually reduce the spatial dimensions to $8 \times 8 \times 128$. The resulting features are flattened into 8,192 values and passed through a fully connected layer containing 512 outputs. MFM then reduces this output to a compact 256-dimensional facial representation. Finally, a fully connected classifier produces a score for each identity class. The complete flow of the architecture is presented in Figure 1.

According to the original paper, LightCNN-9 contains approximately 5.556 million parameters and requires approximately 1 GFLOP. It produces a 256-dimensional facial representation and requires approximately 67 ms for feature extraction on a single core of an Intel i7-4790 processor. The paper reports 98.80% accuracy on LFW and 93.40% accuracy on YTF under its stated evaluation protocols [1].

The smaller model size, low-dimensional representation, and use of grayscale images make LightCNN-9 a suitable candidate for real-time facial recognition. However, its fully connected layer requires a fixed input size, and the reported processing time depends on the hardware used by the original authors. The LFW result also represents face verification rather than the identity-classification task required by our final system. For our implementation, the final classifier can be replaced with a new classifier for the identities in our own dataset while retaining the earlier layers for facial feature extraction.

LightCNN-9

Facial feature extraction and identity classification

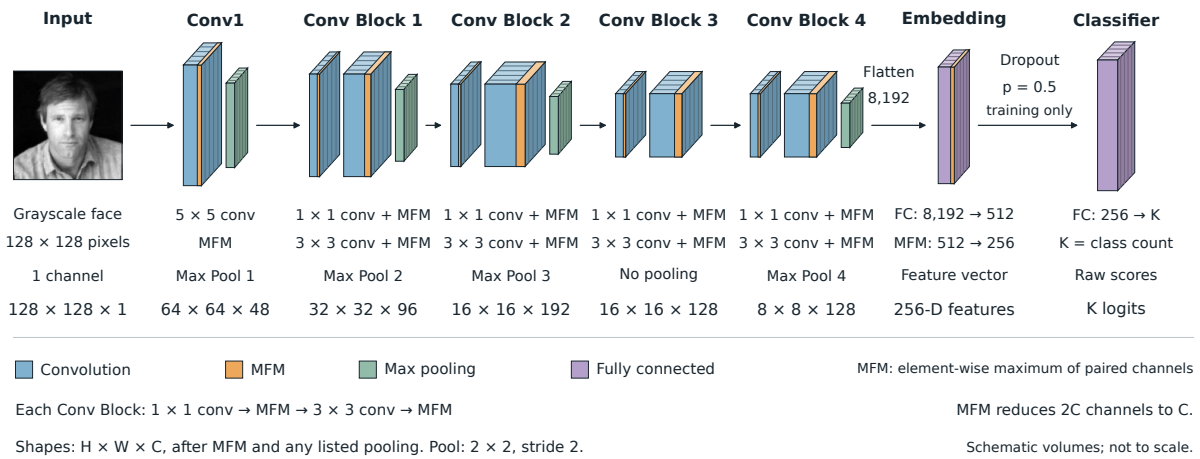


Figure 1: Architecture of LightCNN-9 showing the convolution blocks, MFM operations, pooling layers, 256-dimensional embedding, and identity classifier.

III. Architecture 2: Name

[Write the analysis of Architecture 2 here.]

IV. Architecture 3: Name

[Write the analysis of Architecture 3 here.]

V. Architecture 4: Name

[Write the analysis of Architecture 4 here.]

VI. Analysis and Discussion

The four architectures will be compared using the results reported in their original research papers. LFW is selected as the primary dataset because it is widely used for evaluating facial recognition models. YTF is selected as the second dataset because it evaluates recognition from video frames containing changes in pose, expression, lighting, and image quality. A result will only be treated as a direct comparison when the models use the same dataset, evaluation protocol, and performance metric. Results obtained using different protocols will be discussed separately. Along with recognition performance, the comparison will consider the number of parameters, embedding dimension, inference time, and the hardware used for testing. Table 1 provides the common structure for recording these results. The remaining values will be added after the other architectures have been studied.

VII. Conclusion

[Complete the approximately 100-word conclusion after the remaining architectures and comparison results have been added.]

References

- [1] X. Wu, R. He, Z. Sun, and T. Tan, "A light CNN for deep face representation with noisy labels," *IEEE Transactions on Information Forensics and Security*, vol. 13, no. 11, pp. 2884–2896, 2018.

Table 1: Preliminary comparison of the four facial recognition architectures using results reported in their original papers.

Architecture	LFW Acc. (%)	YTF Acc. (%)	Parameters	Embedding	Inference time
LightCNN-9	98.80	93.40	5.556M	256-D	67 ms ^a
Architecture 2	–	–	–	–	–
Architecture 3	–	–	–	–	–
Architecture 4	–	–	–	–	–

^a Reported for feature extraction using one core of an Intel i7-4790 processor. Accuracy values should only be compared directly when the evaluation protocols are equivalent.